

Autonomous Theory Construction Laboratory

Lecture 8: Reproducing a Converged Search

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Overview

A converged search is reproduced from its environment, data, seeds, and acceptance rule without rerunning open-ended exploration.

1 Problem definition

The scientific question and the admissible output are specified before any search begins.

$$CRS = (D_{ref}, H_{target}, E, template)$$

2 Data and model interface

Data schemas and model interfaces are treated as part of the method, not as post-processing details.

$$score_{CRS} = fit(H, D) + form(PDF)$$

3 Search and evaluation

Proposal and evaluation are kept separate so that a candidate cannot alter its own test.

$$status_{2026} = saturatedregressionset$$

4 Reproducibility

Environment, data version, random seed, and decision record are sufficient to reproduce the reported run.

5 Exercise

The exercise asks for a small implementation and an error analysis rather than a polished narrative.

Checks

- Verify dimensions and sign conventions.
- State the boundary conditions used in each limit.
- Report numerical precision separately from model uncertainty.

Reading

- CRS-26-SAT manifest
- Campus refinement archive operations note
- RLP replay appendix